Total No. of Questions : 8]

SEAT No. :

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T.E. (Computer Engineering)

DISTRIBUTED SYSTEMS

(2019 Pattern) (Semester - I) (Elective - I) (310245 C)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, and Q7 or Q8.***
2) ***Neat diagrams must be drawn whenever necessary.***
3) ***Assume suitable data, if necessary.***

Q1) a) What is clock synchronization? Explain in brief clock synchronization algorithms. **[6]**

b) What is mutual exclusion? Compare centralized and decentralized mutual exclusion algorithms. **[6]**

c) Explain how logical positioning of nodes is done in Gps location system. **[6]**

OR

Q2) a) Explain in detail lamport's logical clock. **[6]**

b) Explain with suitable example how butly election algorithm works. **[6]**

c) Explain gossip-based overlay construction of gossip-based contribution. **[6]**

Q3) a) Explain the following naming system of file system. **[6]**

i) flat naming.

ii) structured naming.

b) Explain file service architecture of distributed file systems. **[6]**

c) Explain with suitable example, Andrew file system. **[5]**

OR

Q4) a) What is attributed based naming? Explain. **[6]**

b) What are identifiers? Explain. **[6]**

c) Explain with suitable example, Suns network file system. **[5]**

P.T.O.

Q5) a) Differentiate between data - centric and client - centric consistency models. **[6]**

b) Describe architecture of replicated data management. **[6]**

c) Explain following terms w.r.t. replica management. **[6]**

i) Content replication.

ii) Content distribution.

OR

Q6) a) What is replication? Enlist reasons for replication. **[6]**

b) Explain eventual consistency model. **[6]**

c) Explain how replicated objects are managed by replica management. **[6]**

Q7) a) What is fault to lerance? Explain in short failure models. **[6]**

b) Explain failure masking in distributed system. **[6]**

c) Describe check pointing for recovery. **[5]**

OR

Q8) a) Describe RPC semantics in presence of failures. **[6]**

b) Explain failure models in distributed system. **[6]**

c) Describe atomic multicast for reliable group communication. **[5]**

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